Question 1 (100 points) ✓ *Saved*

1. **[15 points]** A research team has 8 statisticians, 6 computer scientists, and 5 engineers. A committee of 7 people is selected at random.

   Find the probability that the committee contains at least one person from each group and that no group has more than 3 people on the committee.

2. **[20 points]** A bag contains three biased coins. Coin 1 has probability 0.25 of landing heads, Coin 2 has probability 0.[illegible] Exam 1 [illegible]nding heads, and Coin 3 has probability 0.75 of landing heads.

   One coin is selected at random and tossed four times. Let X be the number of heads observed.

   (a) Find $P(X = 3)$.

   (b) Given that $X = 3$, find the probability that Coin 3 was selected.

3. **[10 points]** Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} \frac{3}{8}x^2, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

   Calculate

$$\mathrm{Var}(3X^2 - 2).$$

4. **[15 points]** Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} \dfrac{c}{x^5}, & x > 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) **[7 points]** Find the value of c.

(b) **[8 points]** Find $P\left(|X-1| < \frac{1}{2}\right)$.

5. **[20 points]** A random variable Y is generated using the following two-stage procedure. First, a machine is selected. Machine 1 is selected with probability 0.35, and Machine 2 is selected with probability 0.65. Second, after the machine is chosen, generate Y from that machine's density.

If Machine 1 is selected, then Y has density

$$f_1(y) = 2y, \qquad 0 < y < 1.$$

If Machine 2 is selected, then Y has density

$$f_2(y) = 6y(1-y), \qquad 0 < y < 1.$$

(a) **[12 points]** Find $P(Y > 0.6)$.

(b) **[8 points]** Given that $Y > 0.6$, find the probability that Machine 2 was selected.

6. **[20 points]** Let A, B, and C be three events such that

$$P(A) = 0.45, \qquad P(B) = 0.50, \qquad P(C) = 0.40,$$

$$P(A \cap B) = 0.25, \qquad P(A \cap C) = 0.18, \qquad P(B \cap C) = 0.20,$$

and

$$P(A \cup B \cup C) = 0.82.$$

(a) **[15 points]** Find $P(A \mid B \cup C)$.

(b) **[5 points]** Are A and $B \cup C$ independent? Justify your answer.